Kerstin (Yu Han) Sun

Houston, TX · 510-996-6090 · kerstinsun2000@gmail.com · linkedin.com/in/yuhanssun · github.com/yuhansun33

AI Engineer Intern at NVIDIA and former Data Analyst at Amazon, specializing in agentic AI systems and full-stack engineering. Delivered an end-to-end automation platform that cut operational cycle time by 98% with agentic AI workflows and backend design. Currently pursuing an M.C.S. at Rice University.

Work Experience

NVIDIA — Santa Clara, CA
AI Engineer Intern — *May 2026 – Jul 2026*

- **Full-Stack AI Platform:** Built and shipped a production application that reduced semiconductor BOM creation time from **16 hours** to **15 minutes**, enabling self-serve for operations and ATG engineers
- **Agentic Execution Workflow:** Engineered a tool-calling AI agent that decomposes natural-language requests into dependency-ordered execution flows across **10** backend tools, gating writes behind human confirmation
- **Deterministic Correctness Layer:** Designed a backend engine that offloads part-identifier generation from the LLM to guarantee zero-hallucination accuracy across **13+** chip-assembly process types, with **90%+** unit test coverage

Amazon — Taipei, Taiwan
Data Analyst — *Jul 2024 – Jul 2025*

- **Segmentation model:** Cut new seller launch time from **142** to **60** days with a K-Means segmentation model that prioritized high-potential onboarding targets, validating profile separation with ANOVA before rollout
- **ETL pipeline:** Designed a modular Python/SQL pipeline consolidating multi-source seller data and remodeled the reporting layer, cutting report latency **40%** for **5** stakeholder teams
- **Marketing analytics:** Built a marketing analytics framework, delivering **30+** data-driven seller consultations that contributed to a **13%** increase in ad spend and improved campaign ROAS

Syntec Technology — Hsinchu, Taiwan
Data Engineer Intern — *Jun 2023 – Nov 2023*

- **Data integration:** Built an SAP ERP ingestion pipeline with automated validation checks feeding cross-department production dashboards for **50+** stakeholders, and automated manual packaging workflows to raise line throughput **10%**

Education

Rice University — Houston, TX
M.C.S., Computer Science | GPA: 4.0/4.0 — *Aug 2025 – Dec 2026 (Expected)*

National Yang Ming Chiao Tung University — Hsinchu, Taiwan
B.S. in Industrial Engineering | GPA: 3.97/4.0 — *Sep 2020 – Jun 2024*

Projects

Prompt-Aware Image-Token Pruning for Multimodal Inference | *PyTorch, LLaVA, CLIP ViT* — *Jan – May 2026*

- Distilled LLaVA-1.5-7B's attention into a **1.3M**-parameter scorer that prunes image tokens before the projector, running **150**× faster than a GradCAM scorer and cutting attention compute **10**×
- Matched full-model accuracy on TextVQA at **25%** token retention, **12.3 points** above random pruning, while deciding token relevance in **0.43 ms** per image

Zero-Shot Job Shop Scheduling with Degree-Normalized GNNs | *PyTorch, GNN, RL* — *Jan – May 2026*

- Found why a scheduling model trained on 10-machine layouts collapsed on 15-machine ones: summing neighbor signals let internal strength balloon with scale. Averaging instead cut cross-size error from **69.8%** to **22.3%** and training time **3**×

Global Content Trending Analysis Platform | *Kafka, PySpark, BigQuery, AWS EC2, Docker* — *Dec 2025 – Jan 2026*

- Built a cross-cloud streaming pipeline over **150K+** daily events, watermarking stateful aggregations for out-of-order arrival and partitioning/clustering BigQuery tables to cut scan cost; deployed with Docker and GitHub Actions

Technical Skills

Languages: Python, TypeScript/JavaScript, SQL, C/C++

AI & ML: PyTorch, LLM agents & tool calling, NeMo Agent Toolkit, RAG, distillation, GNN, reinforcement learning

Backend & Infra: FastAPI, React, REST API, Docker, Kubernetes, CI/CD, Git, PostgreSQL, MongoDB

Data & Cloud: Kafka, Spark, ETL, BigQuery, Pandas, AWS (S3, EC2, EMR), GCP